

AUTOMATIC ANALYSIS OF GAMMA-RAY SPECTRA FROM GERMANIUM DETECTORS

Hypermet function

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A review is given of techniques for the complete automatic analysis of multichannel gamma-ray spectra from high-resolution germanium detectors, and a FORTRAN computer program is described. Previously reported fitting methods, peak shape functions, and peak search techniques are discussed. A semi-empirical peak shape is presented which gives a good fit to the variety of peak shapes and intensities which may be found in a spectrum. Equivalence is shown between a peak search using the covariance of a general peak shape with the data and a search using a general zero-area transform on the data. A comparison is made of the sensitivities of four different transforms when used in a peak search.

1. Introduction

The computer program HYPERMET has been developed at the Naval Research Laboratory for automatic and efficient analysis of multichannel pulse-height spectra from high-resolution germanium gamma-ray detectors. With resolutions of the order of 0.1% for gamma rays in the MeV energy range, a typical spectrum may have several tens of gamma rays easily recognizable above the continuum, with intensities varying by many powers of ten. These will be mostly well resolved, but occasional close-lying or overlapping peaks will occur and must be dealt with in the analysis. This complexity precludes hand analysis, and with a number of such spectra to process the spectroscopist has time for little more than to specify for analysis selected regions or the whole of each spectrum to be processed. The complete job of identifying peaks, setting up fitting parameters, and extracting peak centroids and areas should and can be done more efficiently and consistently by a large high-speed scientific computer.

A number of computer programs for analysis of germanium detector spectra have been reported^{1-7, 14}). A review is given in ref. 8. None of these seemed to satisfy all the requirements of a completely automatic and reliable peak search and fit in an efficient, easy to use program. In the development of program HYPERMET, many techniques and algorithms were considered from refs. 1-8. We have adopted those which worked best over the wide variety of spectra tested, and have modified or devised new techniques where necessary for efficiency, utility, and reliability.

The philosophy developed for program HYPERMET has been to require as little as possible input from the user to the computer, both to minimize the demands on the user's time and to minimize the

possibility of human error, as for example by a mispunched card or a misplaced parameter. A minimum of only three input cards is required for a fit: (1) a title for the spectrum; (2) one or more cards specifying the limits of regions to be fit and a rough estimate of the peak width; and (3) a blank card indicating the end of analysis of the spectrum.

In addition, many optional features, including an energy calibration, Calcomp plot and input-output variations are available to the user by reading additional cards with the option specified by name. Other optional cards allow the user to fix the values of variable parameters or to change the values of initialized parameters. Once optimum values are found for a given system there should be little need to change parameters from run to run. Thus the program retains great flexibility, while at the same time remaining simple and straightforward for routine use.

2. Principles of operation

The main program HYPERMET is written in a structured, modular form. The data is read in, and then various subroutines are called in required order to carry out the analysis specified by the user. A schematic flow chart of the main program is given in fig. 1, which shows the relationship of the program steps and subroutines referred to in this and the following sections.

After reading in the data and the first region card, a peak search is made to determine the locations of any peaks within the specified region. The search, as described in sect. 3, is completely automatic with a high probability of finding all true peaks while discriminating against statistical fluctuations, Compton edges, and other spurious or non-peak features of the spectrum. When a peak is found, a search is made for

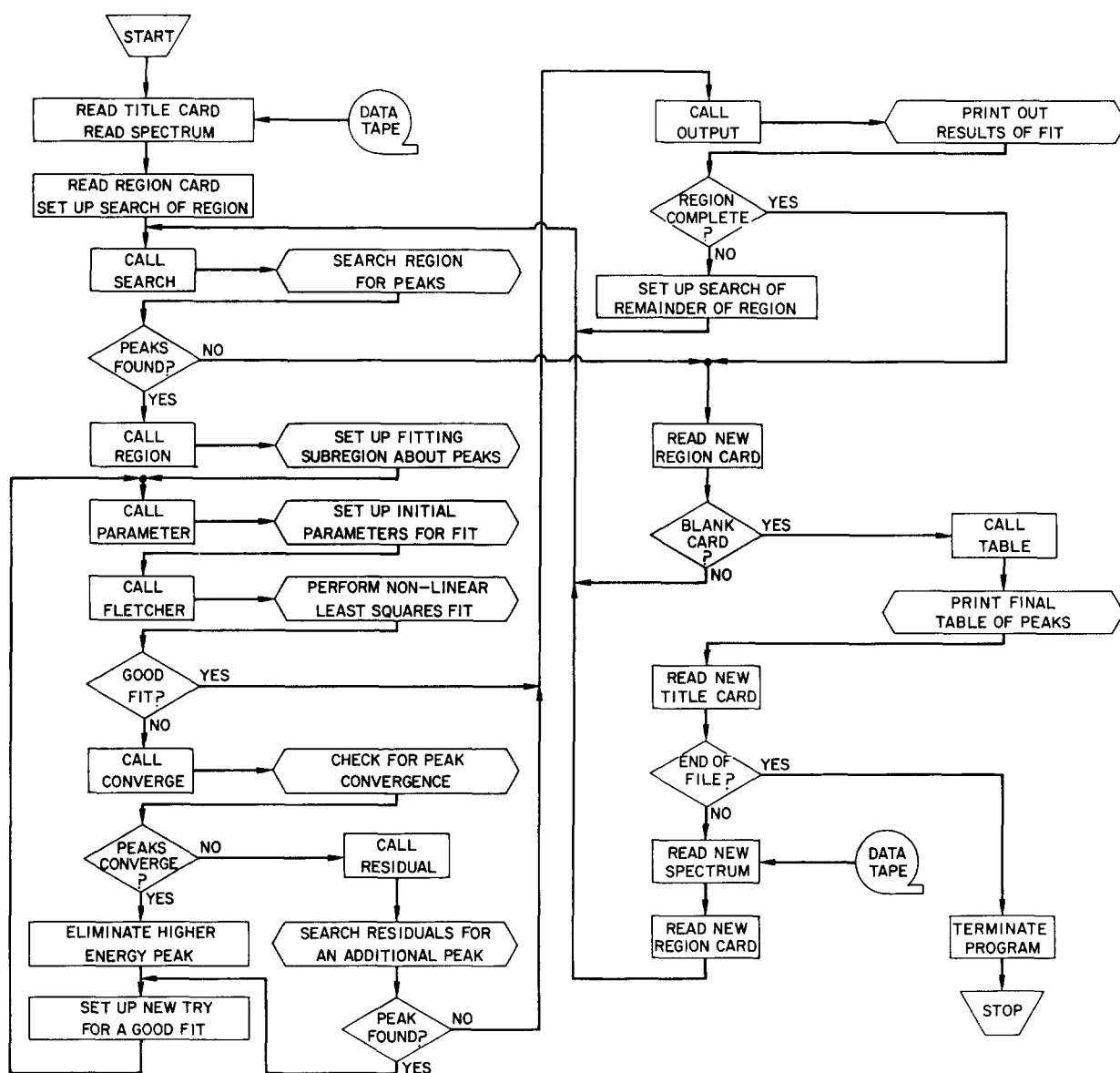


Fig. 1. Schematic flow-chart for program HYPERMET. Rectangular blocks represent actions taken in the main program, and hexagonal blocks identify steps processed by a subroutine. Decision branches are indicated by a diamond.

any additional close-lying peaks which could distort the fit if not included. A fitting subregion is then defined about the peak or peaks. Within this subregion initial estimates are made for the parameters describing the background and peak shape to use as starting values for the fit. The semi-empirical parameterized peak shape function used in the fit is described in detail in sect. 4. It has been developed to be both complete enough and flexible enough to give a satisfactory fit to the variety of peak shapes, sometimes overlapping, that may be found in a spectrum,

including the qualitatively different shapes of full energy, single escape, and double escape peaks.

An iterative non-linear least-squares fit, discussed in sect. 5, is then performed for optimum values of the peak shape and background parameters using a variable-metric fitting routine by Fletcher^{9,10}). Various tests described in sect. 6 then are made to evaluate the fit. If a good fit is obtained, the final parameters and resultant fit are printed out. If not a good fit, attempts are made to improve the fit; a check is made for the possible convergence of two or more peaks, and if so

all but the lowest energy of these are eliminated, and the fit is repeated. Next the residuals are searched for the largest statistically weighted value of data minus fit above a given threshold, and if found a new fit is made with an additional peak at the position of the largest such residual. In this way even very weak but statistically significant peaks on the shoulder of a strong peak are detected and analyzed. If the residuals are all small and no peaks have converged to within the uncertainty in their positions, then the results of the fit are printed out. When all regions of the spectrum have been searched for peaks, a final summary output table is printed of each peak's position, width, area, and count rate and the standard deviations of each of these values. These parameters may also be punched out on cards for input to other programs for further data analysis.

3. Peak search routine

The apparently simple task, to the eye, of selecting narrow peaks that rise significantly above a smoothly varying continuum is not straightforwardly programmed on a computer. The problem is first of defining the continuum and then of distinguishing true peaks from statistical fluctuations and background features such as Compton edges. One method that has been used¹⁾ is to define the continuum iteratively by alternately smoothing the data and truncating remaining features at a level of a few standard deviations. At best, this can be consuming of computer time.

A less obvious, intuitively, but mathematically more rigorous method of minimizing the effects of the continuum during the peak search is to calculate a smoothed second difference of the data; and this procedure is very straightforward to program efficiently. A measure of the sharpness or amount of curvature in the data is given by the second difference or change in slope of a smooth curve drawn through the data. If the smoothing is done by averaging over a distance roughly comparable to the peak width, then this function acts as a filter. It is relatively transparent to slowly varying background features and high frequency statistical noise, but is very sensitive to features which vary sharply over a range of the order of the defined peak width. Various types of smoothing functions have been proposed^{2, 11-13)}. Routti and Prussin²⁾ have used the second derivative of a Gaussian. Mariscotti¹¹⁾ has proposed multiple smoothing functions, which after several smoothing iterations also begin to look much like the second derivative of a Gaussian. Recently Blok et al.¹⁴⁾, have proposed

calculating the covariance of the data with a Gaussian shape.

All of the transforms given above can be classified as belonging to the general group of symmetric zero-area transforms. The expected value of these transforms on a linear background is zero, but statistical fluctuations can produce a more or less peak-like shape. The more peaked the transform is itself, the more peak-like will be the transform of a narrow one or two channel fluctuation. The simple square-wave-like shape described below seems best to filter out this high frequency noise as well as the low-frequency background continuum. However, the peaked transforms may be somewhat better in enhancing a true peak over a statistical fluctuation with a width of the order of the true peak width.

In subroutine SEARCH of program HYPERMET the negative smoothed second difference, defined so as to be positive at a peak, is calculated by a sliding transform, thereby saving storage space since it not necessary to store a complete smoothed spectrum. For an integer width M , we define

$$\begin{aligned} \text{KON}(J) &= \sum_{I=J}^{J+M-1} [K(I) - K(I-M)] - \\ &\quad - [K(I+M) - K(I)] \\ &\equiv \sum_{I=J-M}^{J+2M-1} L(I) * K(I), \end{aligned}$$

where

$$\begin{aligned} K(I) &= \text{number of data counts in the } I\text{th channel, and} \\ L(I) &= -1 \quad \text{for } J-M \leq I \leq J-1, \\ &= 2 \quad \text{for } J \leq I \leq J+M-1, \\ &= -1 \quad \text{for } J+M \leq I \leq J+2M-1. \end{aligned}$$

The coefficients $L(I)$ define a square-wave-like, zero-area convolution function which effects a transform from the data $K(I)$ to the negative smoothed second difference $\text{KON}(J)$. This transform is then shifted over the data by varying J . If M is approximately equal to the full width at half maximum (fwhm) of the peaks, then $\text{KON}(J)$ is large and positive at a peak centered between J and $J+M-1$. For a slowly varying background, the expected value $\langle \text{KON}(J) \rangle$ is approximately zero because of the zero-area of the transform L .

Since the statistical variance in the counts $K(I)$ is just $K(I)$ itself, the variance in $\text{KON}(J)$ is given by

$$\text{VAR}(J) = \sum_{I=J-M}^{J+2M-1} L(I) * L(I) * K(I).$$

For a slowly varying background region, replacing $K(I)$ by its average value $\bar{K}$ gives

$$\langle \text{VAR}(J) \rangle = 6M\bar{K}.$$

Since the standard deviation is the square root of the variance, the function

$$\text{CONV}(J) = \text{KON}(J)/[\text{VAR}(J)]^{1/2}$$

is normally distributed about zero with unit standard deviation, *except in the vicinity of a peak or other sharply varying feature within a range of the order of the width M .*

A threshold value of 4.0 standard deviations has been found to give sufficient sensitivity to detect almost all valid peaks and yet discriminate against statistical fluctuations in the data. A threshold value of 3.0 will detect even very weak peaks at the risk of detecting some spurious peaks due to statistical fluctuations.

The most prominent background feature in gamma-ray spectra is usually the Compton edge that begins below the peak at the highest energy capable of being deposited by a gamma ray which is Compton-scattered out of the detector. The shape of the smoothed second difference is quite different for a peak than for a Compton edge and could be used to discriminate against the latter¹¹). However, for spectra taken with a high-resolution germanium detector the width of the onset of this feature is sufficiently greater than the fwhm of the peaks, that the function $\text{CONV}(J)$ is found to be insensitive to Compton edges without further shape discrimination.

The peak search begins from the low-energy end of the specified region of the spectrum. Once a peak has been detected in subroutine SEARCH the channel with maximum counts is located, tests are made for a partially resolved doublet, and a search is made for nearby peaks which may be close enough to interfere with an accurate background determination. In subroutine REGION a subregion about the peak or peaks is defined, large enough to permit a reasonable background determination but small enough to conserve fitting time. After the fit, subroutine SEARCH is reentered to search for additional peaks beyond the subregion just fitted.

Recently Blok et al.¹⁴) have demonstrated that a channel-by-channel calculation of a weighted covariance of the data with a Gaussian-shaped function, or in fact any general peak-shaped function, will have similar results to the peak-search method discussed above. The covariance method was proposed¹⁴) as a

means of detecting doublets, but at this it was successful only if they were separated by the order of a full peak width. Thus, it did not eliminate the necessity of a residual search¹⁴). Moreover the expression as given appeared to require, for an n -channel transform, several times n floating-point multiplications and additions per channel and thus to use a large amount of computation time compared to the simple fixed-point summation described above. However, the idea is of interest as a general means of searching for any peak shape as long as it is roughly symmetric.

It is easy to show that the covariance of any symmetric function with a linear background is zero. The general definition of the covariance of a function C with data K centered at channel J is given by

$$\text{COV}(K, C) = \frac{1}{n-1} \sum_{I=J-W}^{J+W} [K(I) - \bar{K}] \cdot [C(I-J) - \bar{C}],$$

where

$$\bar{K} = \frac{1}{n} \sum_{I=J-W}^{J+W} K(I),$$

$$\bar{C} = \frac{1}{n} \sum_{I=-W}^W C(I),$$

and the total width of the function C is $n = 2W + 1$. If K is linear in the interval from $J-W$ to $J+W$ then its mean $\bar{K}$ occurs at the midpoint J , and the function $K(I) - \bar{K}$ is antisymmetric about J . However, if C is symmetric so is $C(I) - \bar{C}$, and the dot product of any symmetric function with an antisymmetric function is zero. Thus the covariance of any symmetric function with a linear background has an expected value of zero.

The above definition of the covariance reveals another interesting property. Since the $C(I)$ do not vary as J varies over the data the covariance expression can be rewritten as a sliding transform by observing that the second term in the product is zero,

$$\begin{aligned} & \frac{1}{n-1} \sum_{I=-W}^W \bar{K} \cdot [C(I) - \bar{C}] \\ &= \frac{1}{n-1} \bar{K} \left(\sum_{I=-W}^W C(I) - \sum_{I=-W}^W \bar{C} \right) \equiv 0. \end{aligned}$$

Therefore

$$\text{COV}(K, C) = \frac{1}{n-1} \sum_{I=J-W}^{J+W} K(I) \cdot [C(I-J) - \bar{C}].$$

Thus the transform

$$C(I)' = \frac{1}{n-1} [C(I) - \bar{C}]$$

is an equivalent definition of the covariance and belongs to the general group of symmetric zero-area transforms.

The use of a zero-area gaussian-shaped transform to calculate the covariance requires either floating point arithmetic or the approximation of the function by an integer function normalized to a peak amplitude of perhaps 100. This is also necessary for a transform using the second derivative of a gaussian. Thus these calculations are more complex and require more computation time than for a simple square-wave-like transform. An alternative for achieving increased sensitivity of signal over noise has been given by Robertson et al.¹²⁾, in which the width of the negative tails of the square-wave-like transform is increased relative to the width of the positive central portion. Likewise the height of the central portion is increased to maintain zero net area. Again the computation time is increased but only in proportion to the total width of the transform. The improvement in signal to noise is the result of a better averaging of the background on either side of the peak due to the broader negative tails of the transform.

A comparison was made of the following four transforms:

- Square wave, central width $M = 5$, amplitude $A = 2$, tail width $T = 5$, total width $= 3M = 15$.
- Rectangular wave, $M = 5$, $A = 4$, $T = 10$, total width $= 5M = 25$.
- Covariance, calculated using a zero-area Gaussian, $C - \bar{C}$, where $C = \exp \{-(I-J)^2/D^2\}$, width $D = 2.5$, fwhm $= 4.16$, total width $= 25$.
- 2nd derivative of a Gaussian with an optimum width of $D = 4.2$, total width $= 25$.

For a general transform C of total width $2W+1$ with a unit amplitude peak shape Y centered at J , we can define a filter strength

$$F(Y, C) = \frac{\left[\sum_{I=J-W}^{J+W} Y(I) * C(I-J) \right]^2}{\sum_{I=-W}^W C(I)^2}.$$

For a peak with amplitude P on a background of height B , the signal-to-noise ratio is defined by $F * P^2/B$. The relative improvement in signal amplitude to noise amplitude is given by $\sqrt{F}$. Table 1 gives the values of F and $\sqrt{F}$, for the transforms (a) to (d) defined above, on a Gaussian peak shape Y with

TABLE 1

Comparison of zero-area transforms.

Transform	F	$\sqrt{F}$	G	$R(G)$	S	$R(S)$
(a) Square wave	1.56	1.25	1.40	3.7 ± 0.2	1.34	3.4 ± 0.4
(b) Rectangular	2.06	1.43	1.77	2.1 ± 0.3	1.44	1.9 ± 0.2
(c) Covariance	2.35	1.53	1.97	1.4 ± 0.2	1.42	1.7 ± 0.4
(d) 2nd derivative	2.00	1.41	1.71	2.7 ± 0.4	1.38	3.0 ± 0.2
(e) Self	3.13	1.77				

$D = 2.5$. Also given is the value of the self-transform

$$F(Y, Y) = \sum Y(I)^2,$$

which is an upper limit for the filter strength F .

Column 4 of the table shows the results of a comparison of the four transforms on computer generated data consisting of a Gaussian peak, with parameter $D = 2.5$, with amplitude 30, or three standard deviations above a constant background of 100. Added to these data was a noise term randomly selected from a normal distribution with unit standard deviation, then multiplied by the expected statistical standard deviation (i.e. the square root of the data counts) and rounded to the nearest integer. The result was a true peak plus one or more noise peaks due to statistical fluctuations. Column 4 of the table shows, for seven sets of data, the average ratio G of the transformed true peak to noise peak amplitudes divided by the amplitudes for the raw data. The result is the relative improvement in the true-peak-to-noise-peak amplitudes. Because only relatively strong noise peaks were included, these values of G are somewhat larger than $\sqrt{F}$ in column 3, but the relative agreement is good. As an indication of the degree of significance of these numbers, the transforms were ranked from 1 to 4 according to G for each data set. Column 5 gives the mean rank $R(G)$ and its standard deviation for the seven data sets.

Column 6 gives the mean sensitivity or signal enhancement S (true peak amplitude of the transformed data divided by the peak amplitude for the raw data), and column 7 shows the mean and standard deviation for the seven data sets of the rank $R(S)$ with respect to S . There is less variation in S than there is in G , indicating that the difference in the signal to noise ratios for these four transforms is due more to noise suppression than to signal enhancement.

Thus, at a threshold level of 4.0 standard deviations above background, the simple square-wave transform, which requires the least calculational effort, should be more than adequate in both sensitivity and signal to

noise. However, if one is routinely interested in all peaks as low as 3.0 standard deviations, table 1 shows that transform b or possibly even (c) might be worthwhile to reduce the time wasted in the detection and analysis of spurious noise peaks. [Transform (d) is not significantly different from (b) in signal-to-noise and requires as much computation as does (c).] The peak search routine in program HYPERMET has been modified to switch from transform (a) to transform (b) whenever the threshold level is set below 4.0 standard deviations.

4. Peak shape parameters

A wide variety of peak-shape functions have been proposed to use in fitting peaks from germanium detectors^{1-8, 14-16}). A review and comparison by fitting experimental peak shapes is given in ref. 16. In general, the proposed shapes fall into two categories, those which are a sum of several functional terms such as a Gaussian peak shape, tail, etc., and those which use two or more distinct functions in the region of the peak and of the tail, joining these smoothly at some point.

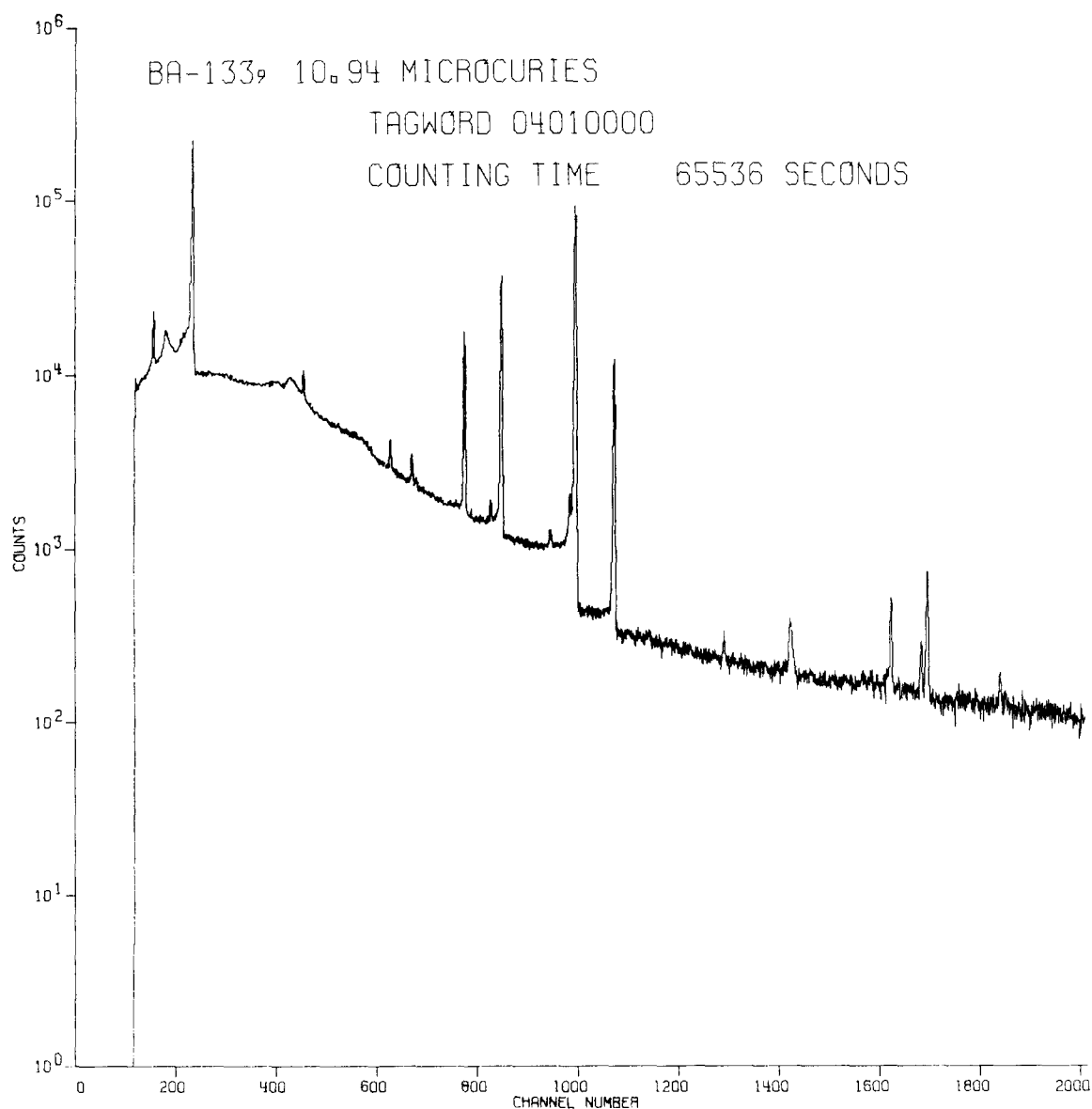


Fig. 2. Example of a spectrum taken with a high resolution germanium detector of the gamma radiation from a ^{133}Ba source. The step or discontinuity at the peak and the tail below the peak are evident on several of the strong peaks.

Most of these proposed shapes are ad hoc and often result in parameters such as joining widths with little or no physical interpretation. The shapes which seem to fit the data best all have a Gaussian peak with the addition of a skewed term or tailing on the low energy side, plus a long tail or step below the peak. Fig. 2 shows a spectrum taken with a germanium detector of a ^{133}Ba source. The strong peaks exhibit many of the peak-shape characteristics discussed below. Particularly noticeable is the discontinuity in the background at the prominent peaks.

In HYPERMET, a semi-empirical parameterized function is used to represent the detector response in the vicinity of a peak. The peak shape plus background is represented by a sum of terms as illustrated in fig. 3. These are discussed below in order of decreasing importance.

1) For an ideal detector, a gamma ray with a negligible natural width would start out as a sharp line and be broadened by statistical and electronic noise into a Gaussian shape

$$\Gamma \exp\{-(x/\delta)^2\},$$

with amplitude Γ and width δ , at position x with respect to the peak position P . For a multichannel spectrum we can let $x = J - P$ to obtain an expression for counts in channel J . The width δ varies approximately as the gamma-ray energy squared.

2) Effects of incomplete charge collection and pile-up can remove counts from the sharp peak, producing in an ideal detector an exponentially decaying distribution below the peak, $\exp(x/\beta)$ for $x < 0$, i.e., for $J < P$. Folding this in with a Gaussian noise of width γ centered at x , gives

$$\int_{-\infty}^0 \alpha \exp(t/\beta) * \exp\{-(t-x)^2/\gamma^2\} dt \\ = \alpha \exp(x/\beta) * \frac{1}{2} \operatorname{erfc}(x/\gamma + \gamma/2\beta),$$

where α is a normalizing amplitude and the complementary error function, $\operatorname{erfc}(u)$ is normalized so that

$$\frac{1}{2} \operatorname{erfc}(u) = 1 \text{ for } u \ll 0, \frac{1}{2} \text{ for } u = 0, \text{ and } 0 \text{ for } u \gg 0,$$

thus providing a smooth cutoff centered at $x = -\gamma^2/2\beta$. This offset results from integration of a Gaussian multiplied by an exponential. The slope β and amplitude α vary with energy. For a good detector, these tend to fall in the range of 0.1–1.0 times the respective peak width δ and amplitude Γ . The noise width γ is usually set equal to δ . The addition of terms 1 and 2 results in a peak with a roughly gaussian shape but skewed or broadened on the low energy side.

3) For very strong peaks one can observe a much longer exponential tail below the peak, due perhaps to surface effects. This is usually reduced two or three orders of magnitude in amplitude τ relative to Γ and has a slope ν one or two orders larger than β . Again folding with a Gaussian noise term of width μ gives

$$\tau \exp(x/\nu) * \frac{1}{2} \operatorname{erfc}(x/\mu + \mu/2\nu).$$

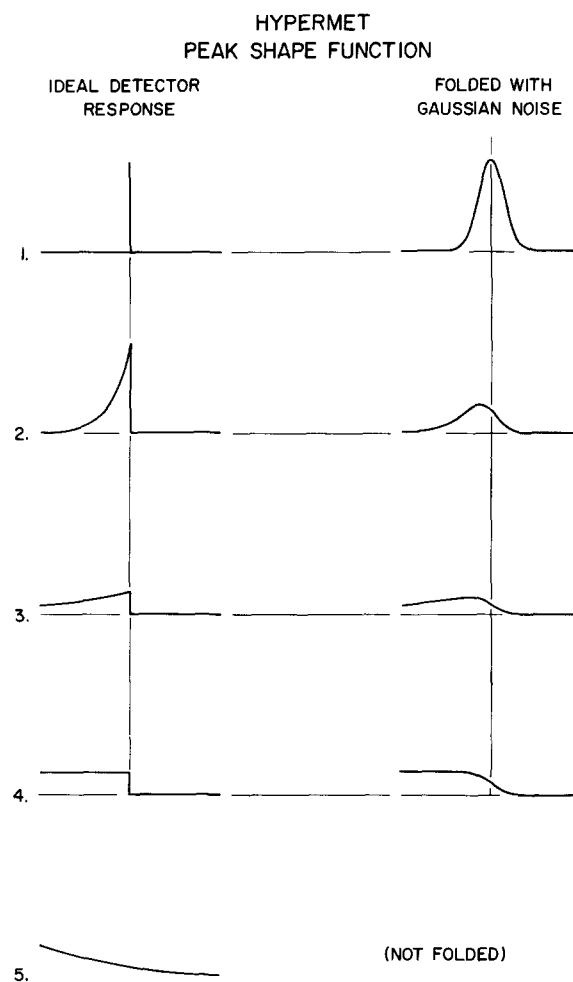


Fig. 3. Illustration of terms used in the fitting function for program HYPERMET. The numbers at the far left refer to the mathematical function for each term as given in the text. The figures at the left indicate the shape of each term for a noiseless "ideal" detector. Terms 1–4 are each associated with a source of discrete energy gamma rays, producing a peak at the position shown by the light vertical line. The amplitudes of terms 2, 3, and 4 relative to term 1 are exaggerated for illustrative purposes. Term 5 represents the background continuum which may be large or small compared to the other terms. At the right is shown the results of folding in a gaussian noise term with terms 1–4. Again the peak position is shown by the light vertical line. The background continuum is not folded.

4) For gamma rays with energy greater than about 2 MeV, electron-positron pair production is an important mechanism of energy transfer to the detector. This leads to positron annihilation resulting in two 511 keV gamma rays, one or both of which can escape from the detector. Thus a source of single-energy gamma rays will produce a full-energy peak, a single-escape peak 511 keV lower in energy, and a double-escape peak 1022 keV lower in energy. These can have significantly different shapes. In addition the annihilation gamma rays can Compton scatter in the detector with the energy given up adding to that of the escape peaks. Summing effects such as this can give the appearance of a step or discontinuity in the background above or below the peak. For lower-energy gamma rays, Compton scattering into the detector from surrounding materials can also produce a significant step or ledge below the peak. A step of amplitude Σ , folded with a Gaussian noise of width σ , gives

$$\Sigma * \frac{1}{2} \operatorname{erfc}(x/\sigma).$$

5) Finally, the background term for the continuum within a few fwhm on either side of a peak is usually well represented by a first or second order polynomial

$$B(J) = a + b + cJ^2.$$

Terms 3 and 4 add to the continuum term 5 to give a background with a sharp step at the peak and which falls off slowly below the peak.

For a given energy gamma ray all these effects except the background continuum are proportional to the peak amplitude; $\alpha = A\Gamma$, $\tau = T\Gamma$ and $\Sigma = S\Gamma$. Theoretically and in practice the cutoff widths should usually equal the peak width; $\gamma = \mu = \sigma = \delta$. If there are m peaks in the subregion to be fit, the fitting function for channel J becomes

$$Y(J) = \sum_{I=1}^m \Gamma(I) * (\exp\{-[x(I, J)/\delta(I)]^2\} + \quad (1)$$

$$+ A \exp\{x(I, J)/\beta\} * \frac{1}{2} \operatorname{erfc}\{x(I, J)/\gamma + \gamma/2\beta\} + \quad (2)$$

$$+ T \exp\{x(I, J)/\nu\} * \frac{1}{2} \operatorname{erfc}\{x(I, J)/\mu + \mu/2\nu\} + \quad (3)$$

$$+ S * \frac{1}{2} \operatorname{erfc}\{x(I, J)/\sigma\} + \quad (4)$$

$$+ B(J), \quad (5)$$

where $x(I, J) = J - P(I)$ for the I th peak located at position $P(I)$.

This function is similar to the one used by Varnell and Trischuk³, but with the addition of the step (term 4). These authors and others have found it necessary to add an additional term similar to term 2, but with x replaced

by $-x$, due to tailing on the high energy side of the peak. If this occurs in the spectrum it is usually due to an improperly adjusted pole-zero in the main amplifier, an excessively high count rate and/or insufficient baseline restoration. It is best not to rely on the program to correct for electronic instrument problems since adding these additional parameters would further complicate the fit. In the few spectra where this has been a problem, satisfactory fits were obtained, but a small additional peak was added on the high side of each strong peak to account for the high-energy tailing.

5. Method of optimization

The expression for the fitting function $Y(J)$ above is very non-linear and has a formidable number of parameters. For low-energy peaks it may be possible to express the variation of many of these parameters as a simple function of energy^{1,3,14}). This could reduce the number of free parameters needed in the fit. However, this introduces the problem of either performing a good calibration and thereafter maintaining a stable system for each spectrum measured or of obtaining a new calibration for each spectrum, many of which may not have enough strong isolated peaks for a good calibration. Moreover, for energies of about 2 MeV or above the values of the parameters turn out to be significantly different for full-energy, single-escape, and double-escape peaks. In HYPERMET, the peak width $\delta(J)$, which is usually slowly varying, and the slopes β and ν are usually constrained to be the same for all peaks within a fitting subregion; $\delta(I) = \delta$, $\beta = B\delta$, and $\nu = V\delta$. The cutoff widths γ , μ , and σ are set equal to δ times a constant (usually 1.0). This leaves two parameters $\Gamma(I)$ and $P(I)$ per peak, plus the width δ , the slopes β and ν , the relative amplitudes A , S , and T , and the background coefficients a , b , and c , for a maximum of $2m + 9$ parameters for m peaks. Except for very strong peaks with good background statistics, S , T , ν and c may be neglected leaving a minimum of $2m + 5$ parameters, which must be varied to give the best fit.

This is still a challenging task for any optimization method, but the problem is greatly eased by providing a good initial value for each parameter as a starting point. This is done in subroutine PARAMETER which tests for inclusion of parameters S , T , ν and c , estimates best initial values, and sets up constraints on

* As the program is routinely used, it may not provide a good fit to a subregion containing a mixture of intense single-escape, double-escape and/or full-energy peaks, because they do not have the same widths. An option permits fitting a different width to each peak in a subregion.

the various parameters, most of which are limited by maximum and/or minimum values relative to their initial values.

These constraints are handled functionally as suggested by Lederer⁸, by setting each constrained parameter equal to a function with an appropriately limited range for one of a set of dummy parameters $X(I)$, e.g.

$$\Gamma = X(1)^2 \text{ implies } \Gamma \geq 0 \text{ for all } X(1),$$

$$A = A^- + (A^+ - A^-) \sin(X(2)) \text{ implies } A^- \leq A \leq A^+$$

for all $X(2)$, etc.

The periodicity of the sine function is no problem for any reasonably sized range, and can be advantageous by providing an automatic turnaround at the boundary. This is preferable to the arctangent, for example, where the argument approaches infinity at the boundary. In such a case the minimization routine can get stuck for many iterations striving to reach the ever-distant boundary.

Optimization of the fitting parameters $X(I)$ is accomplished by minimizing the least-square deviation, chisquare (χ^2), of the data $K(J)$ from the function $Y(J, X)$, defined by

$$\begin{aligned} \chi^2 &= \frac{1}{\text{ND}} \sum_J \text{CHI}(J)^2 \\ &= \frac{1}{\text{ND}} \sum_J [(K(J) - Y(J, X))/K(J)^{\frac{1}{2}}]^2, \end{aligned}$$

where the residuals $\text{CHI}(J)$ are individually normalized to unit standard deviation by dividing by $K(J)^{\frac{1}{2}}$. The overall normalization factor ND is given by the number of degrees of freedom in the fit, which is equal to the number of data channels included in the fit minus the number of free parameters X . If the function Y is a good representation of the data, the expected value of χ^2 is 1 and its standard deviation is approximately $(2/\text{ND})^{\frac{1}{2}}$.

HYPERMET uses the variable-metric minimization method¹⁷⁻¹⁹) with the modifications by Fletcher^{9, 10}) which greatly increase the efficiency of the method. Minimization is performed on the function $F(X) = \frac{1}{2}\chi^2$, which is so normalized that the final $\mathbf{H}$ matrix defined below approximates the variance-covariance matrix of the parameters $X(I)$. An iterative search for a minimum in the function F is begun in the direction D , the negative gradient of F ,

$$D(I) = -G(X, I) = -\frac{\partial F(X)}{\partial X(I)},$$

at each iteration the length of the step or increase in each parameter $X(I)$, is determined by a variable metric $\mathbf{H}(I, J)$,

$$DX(I) = \sum_J \mathbf{H}(I, J) D(J) = -\sum_J \mathbf{H}(I, J) \frac{\partial F(X)}{\partial X(J)}.$$

In the neighborhood of a minimum of $F(X)$, where Taylor's approximation is valid, the matrix $\mathbf{H}$ is approximately the inverse Hessian matrix of the function F , which corresponds to the variance-covariance matrix $\mathbf{V}$,

$$\mathbf{H} \cong \left\| \frac{\partial^2 F(X)}{\partial X(I) \partial X(J)} \right\|^{-1} \equiv \mathbf{V}.$$

To within a factor of χ^2 , the diagonal elements $\mathbf{V}(I, I)$ give the variance in the parameters $X(I)$, and the off-diagonal elements $\mathbf{V}(I, J)$ give the covariance of $X(I)$ and $X(J)$.

Generally, in the variable-metric method a starting value for the matrix $\mathbf{H}$ is chosen as a diagonal matrix, usually $\text{VAR}(I) * \delta(I, J)$, where $\text{VAR}(I)$ is an estimate of the variance or square of the uncertainty in the initial value of $X(I)$. At each iteration of the fit, the matrix $\mathbf{H}$ is updated by a formula originally given by Davidon¹⁸) and modified by Fletcher^{9, 10}) so as to successively approximate the unknown variance-covariance matrix $\mathbf{V}$. At the same time the optimum values of X are successively approximated. At the minimum, if a good fit is obtained, the parameters X are determined to within an uncertainty or standard deviation given by

$$\text{SD}(I) = [\mathbf{V}(I, I) \chi^2]^{\frac{1}{2}}.$$

The standard deviation for a function of the parameters such as the area $\text{AREA}(X)$ of a peak is

$$\begin{aligned} \text{SD}(\text{AREA}(X)) &= \left[\sum_{I, J} \frac{\partial \text{AREA}(X)}{\partial X(I)} \times \right. \\ &\quad \left. \times \frac{\partial \text{AREA}(X)}{\partial X(J)} \mathbf{V}(I, J) \chi^2 \right]^{\frac{1}{2}}. \end{aligned}$$

The main advantages of the variable-metric method are that it converges rapidly near the minimum and requires a relatively small amount of additional storage. The only arrays needed are several vectors of length N , and the $N * N$ matrix $\mathbf{H}$, where N is the number of free parameters X . Other non-linear least squares methods^{20, 21}) may require storage of two or even three dimensional arrays with one dimension being the number of data points, which is usually much greater than N .

A major disadvantage of the variable-metric method, as formulated by Davidon¹⁸), is that it often must make a number of tests and trial steps at each iteration in order to assure convergence, thus increasing the computation time required for a fit. This problem is greatly reduced by the modification due to Fletcher^{9, 10}) which is used in HYPERMET. A number of comparisons were made on a variety of data. Although the total number of iterations required for a fit was comparable, Davidon's method required on the average greater than two trial steps per iteration, whereas Fletcher's method generally took an average of two trial steps per iteration for the first N iterations, but only slightly more than one step per iteration thereafter. Since the number of

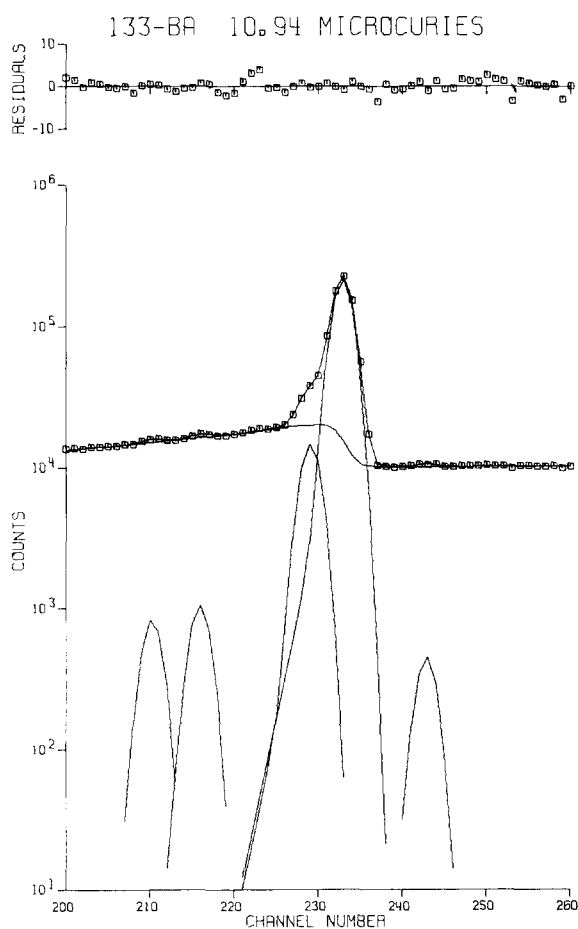
iterations required for a fit is usually several times N , the savings in computing time for this part of the program thus approaches 50%. The overall savings turned out to be between 30% and 40% in comparisons on the same data.

6. Evaluation of the fit

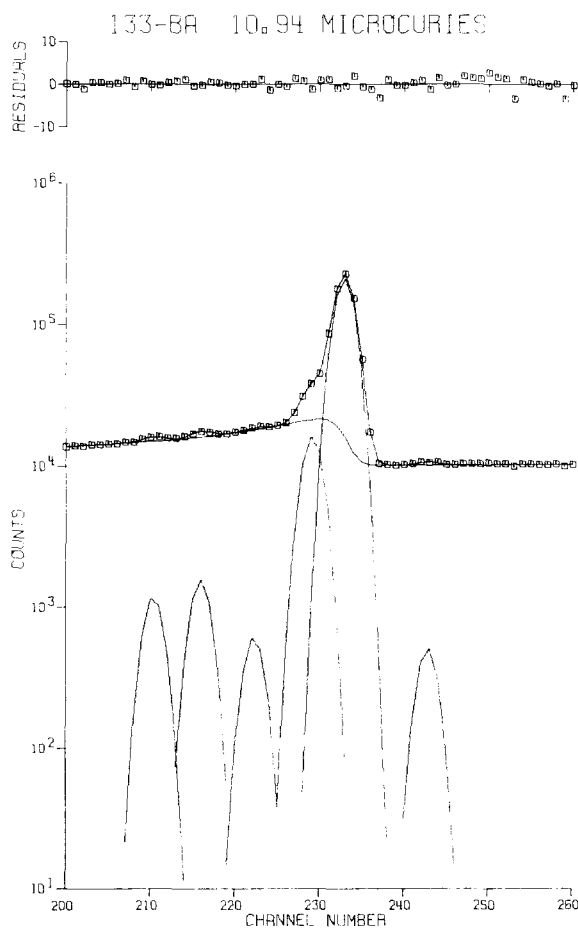
After the optimization several tests are made to determine if the fit is acceptable, or if not, whether the fit can be improved.

a) If this is not the first try, χ^2 is compared to its previous value. If χ^2 has not improved, the results of the previous try are restored and printed out.

b) If χ^2 does not exceed its expected value of 1 by



(a)



(b)

Fig. 4. Example of a fit to a complicated region showing the 80–81 keV doublet of ^{133}Ba , and weak Pb and Bi K X-rays from the decay chain of thorium in the Laboratory background radiation. Fig. 4a, left, shows the optimized fit with five peaks. In fig. 4b, right, is shown the optimized fit with the addition of a sixth peak at approximately channel 222, as determined from a search of the residuals in the fit at the left. Squares are used to indicate the data and the residuals (at top). The fit is given by the line through the data. Also shown by lines is the breakdown of the fit into the individual peak shapes (sum of terms 1 and 2 of fig. 3) and the background (sum of terms 3, 4 and 5 of fig. 3).

more than a preset amount, usually 4.0 standard deviations, the fit is terminated and the results printed out.

c) Subroutine CONVERGE is called which checks if any two or more peaks have converged to within the uncertainties of their positions. If so, all but the lowest in position of these peaks are eliminated and a new try is called for after which the results of the fit are printed out.

d) If none of the above tests are satisfied, the entry RESIDUAL of subroutine CONVERGE is called, which examines the normalized residuals

$$\text{CHI}(J) = [K(J) - Y(J)]/K(J)^{\frac{1}{2}},$$

for indications of an additional peak which was not included in the fit. The negative smoothed second difference of the residuals is calculated for each channel in the subregion by

$$\text{RCON}(J) = \sum_{I=J-M}^{J+2M-1} L(I) * \text{CHI}(I),$$

where the $L(I)$ represents a zero-area convolution function as defined in sect. 3. Since the $\text{CHI}(I)$ are defined with unit standard deviation, RCON has a standard deviation

$$\text{RSD} = \sum_{I=J-M}^{J+2M-1} [L(I) * L(I)]^{\frac{1}{2}} = (6M)^{\frac{1}{2}}.$$

Near the edges of the subregion, $J < M$ or $J > \text{NR} - 2M + 1$, where NR is the total number of channels included in the subregion, a simple sum is used

$$\text{RCON}(J) = \sum_{I=J}^{J+M-1} \text{CHI}(I),$$

which has a standard deviation $\text{RSD} = M^{\frac{1}{2}}$. If any $\text{RCON}(J)/\text{RSD}$ exceeds a preset value, usually 3.0, and is distinct from the nearest peak by greater than 1.5 channels or the uncertainty in the peak's position, then for the largest such $\text{RCON}(J)/\text{RSD}$ a new peak is added at the position $J + M/2$, and a new try is called for. Only one peak is added at each try because a missing peak in the fit may distort the fitted background sufficiently to cause the appearance of spurious peaks elsewhere in the residuals. This process continues until a good χ^2 is obtained, or until no additional peaks are found from examination of the residuals, or until a preset maximum number of tries is reached.

7. Discussion

The various steps necessary for complete automatic analysis of a gamma-ray spectrum have been discussed

and techniques for implementing them have been reviewed. The procedures developed for program HYPERMET have been presented in detail. The program has been tested on a wide variety of data and is now in routine use at this Laboratory. Fig. 4 shows an example of a fit to a complex region including two strong overlapping peaks and several much weaker peaks.

The strong doublet is well fit by the semi-empirical peak shape function used, and all significant peaks were uncovered by the peak search plus residual search. The figure shows the region of the 80–81 keV doublet from the spectrum of ^{133}Ba (fig. 2). Also seen weakly are background Pb and Bi K X-rays from the decay chain of ^{232}Th . The fit to the five largest peaks is shown in fig. 4a. From examination of the residuals there is an additional peak indicated at about channel 222. The fit with this peak included is shown in fig. 4b. The improvement in the residuals is evident. The improvement in the fit is shown by a reduction in χ^2 from 2.275 for five peaks to 1.517 for six peaks.

Several effects on the fit due to the missing peak at channel 222 can be noted by comparing the plots in figs. 4a and 4b. The background is distorted so that the amplitudes of the fitted peaks at channels 210 and 216 are approximately 20% less in fig. 4a than in fig. 4b. In addition, the peak shapes are distorted causing a large skewness in the fitted peak shapes in the fit for fig. 4a compared to the nearly symmetrical peak shapes in fig. 4b. Also evident in this fit is the need for both a step in the background at the peak and a long tail below the peak. In this case the step has an amplitude S of $(1.08 \pm 0.15)\%$ relative to the peak height, and the tail has an amplitude T of $(4.39 \pm 0.09)\%$ with a slope of (11.5 ± 0.7) times the peak width.

Program HYPERMET is written in FORTRAN 63 for the CDC 3800 computer system and fits into one 32k word bank of 48-bit words except for a few system input-output routines. A revision is planned for the new Texas Instruments Advanced Scientific Computer which will be more nearly compatible with IBM 360/370 FORTRAN. A more detailed description of the program along with a user's manual and a complete listing is given in the appendices of ref. 22.

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